

2(a)	rate of change of displacement or change in displacement/time taken	B1
2(b)(i)	$s = ut + \frac{1}{2}at^2$	C1
	$t = [(2 \times 1.25) / 9.81]^{1/2} (= 0.5048 \text{ s})$	C1
	or	
	$v^2 = u^2 + 2as$ $v_{\text{vert}} = (2 \times 9.81 \times 1.25)^{1/2} (= 4.95)$	(C1)
	$t = [2s / (u + v)] = 2 \times 1.25 / 4.95 (= 0.5048 \text{ s})$	(C1)
	$v = d / t = 1.5 / 0.50(48)$ $= 3.0 (2.97) \text{ ms}^{-1}$	A1
2(b)(ii)	vertical velocity $= at$ $= 9.81 \times 0.5048 (= 4.95) \text{ [using } t = 0.50 \text{ gives } 4.9]$	C1
	velocity $= [(v_h)^2 + (v_v)^2]^{1/2}$	C1
	$= [(2.97)^2 + (4.95)^2]^{1/2}$ $= 5.8 (5.79) \text{ [using } t = 0.50 \text{ leads to } 5.7]$	A1
	direction $(= \tan^{-1} 4.95/2.97) = 59^\circ$	A1
2(b)(iii)	kinetic energy $= \frac{1}{2}mv^2$	C1
	$= \frac{1}{2} \times 0.45 \times (5.8)^2$ $= 7.6 (7.57) \text{ J [using } t = 0.50 \text{ leads to } 7.3 \text{ J]}$	A1

Question	Answer	Marks
2(b)(iv)	potential energy = mgh	C1
	= $(0.45 \times 9.81 \times 1.25)$	A1
	= 5.5 (5.52) J	
2(c)	there is KE of the ball at the start/leaving table or the ball has an initial/constant horizontal velocity or the ball has velocity at start/leaving table	B1

Question	Answer	Marks
----------	--------	-------